

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: OPERATING SYSTEMS

COURSE CODE: CSA04

COURSE OUTCOME COVERED

CO2: Analyze process management and deadlock handling techniques to improve CPU utilization and system performance(BL3-BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 10

Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I V.Hussain (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V.Hussain

Question 1: SJF (Non-Preemptive)

Given

Process Burst Time

P1 10

P2 4

P3 6

P4 2

All arrive at **0 ms**.

Step 1: Order (Shortest Job First)

P4 → P2 → P3 → P1

Gantt Chart

0 2 6 12 22

| P4 | P2 | P3 | P1 |

Waiting Time (WT)

WT = Start Time – Arrival Time

Process WT

P4 0

P2 2

P3 6

P1 12

Turnaround Time (TAT)

TAT = WT + Burst Time

Process TAT

P4 2

P2 6

P3 12

P1 22

Average Waiting Time

$$\frac{0+2+6+12}{4} = 5 \text{ ms}$$

Average Turnaround Time

$$\frac{2+6+12+22}{4} = 10.5 \text{ ms}$$

Interpretation

- Average WT = **5 ms**
- Average TAT = **10.5 ms**
- SJF reduces waiting time compared with FCFS, so CPU performance is better for this workload.

Question 2: Round Robin (Time Quantum = 4 ms)

Given

Process Burst

P1 20

P2 5

P3 13

P4 8

All arrive at time 0.

Gantt Chart

0 4 8 12 16 17 21 25 29 33 37 38 42 45 49

|P1|P2|P3|P4|P1|P2|P3|P4|P1|P3|P1|P3|P1|

Completion Time

Process CT

P1 49

P2 17

P3 45

P4 29

Turnaround Time

(TAT = CT)

Process TAT

P1 49

P2 17

P3 45

P4 29

Waiting Time

$$WT = TAT - \text{Burst}$$

Process WT

P1 29

P2 12

P3 32

P4 21

If Quantum = 10 ms

- Fewer context switches
- Lower scheduling overhead
- Better CPU utilization
- Behaves closer to FCFS
- Interactive response becomes slower

Question 3: Banker's Algorithm**Allocation Matrix****Process A B C**

P1 0 1 0

P2 2 0 0

P3 3 0 2

P4 2 1 1

P5 0 0 2

Maximum Matrix

Process A B C

P1 7 5 3

P2 3 2 2

P3 9 0 2

P4 4 2 2

P5 5 3 3

Need Matrix

Need = Maximum – Allocation

Process A B C

P1 7 4 3

P2 1 2 2

P3 6 0 0

P4 2 1 1

P5 5 3 1

Available Resources

Allocated

A = 7

B = 2

C = 5

Available

A = 10 – 7 = 3

B = 5 – 2 = 3

C = 7 – 5 = 2

Available = **(3,3,2)**

Safe Sequence

Available (3,3,2)

P2 executes → Available = (5,3,2)

P4 executes → Available = (7,4,3)

P5 executes $\rightarrow$ Available = (7,4,5)

P1 executes $\rightarrow$ Available = (7,5,5)

P3 executes $\rightarrow$ Available = (10,5,7)

Safe Sequence

P2 $\rightarrow$ P4 $\rightarrow$ P5 $\rightarrow$ P1 $\rightarrow$ P3

Conclusion

The system **is in a safe state**.

Question 4: Resource Allocation Graph

Text Representation

R1 $\rightarrow$ P1

P1 $\rightarrow$ R2

R2 $\rightarrow$ P2

P2 $\rightarrow$ R3

R3 $\rightarrow$ P3

P3 $\rightarrow$ R1

Deadlock?

Yes.

Cycle:

P1 $\rightarrow$ R2 $\rightarrow$ P2 $\rightarrow$ R3 $\rightarrow$ P3 $\rightarrow$ R1 $\rightarrow$ P1

Hence, **Deadlock exists**.

Recommended Method

Recovery

- Abort one process or preempt resources.
- This breaks the cycle and removes the deadlock.

Question 5: CPU Utilization

Formula

$$CPU = 1 - p^n$$

Given

$$p = 0.8$$

(a) n = 4

$$CPU = 1 - (0.8)^4 = 1 - 0.4096 = 0.5904$$

CPU Utilization = **59.04%**

(b) n = 6

$$CPU = 1 - (0.8)^6 = 1 - 0.262144 = 0.737856$$

CPU Utilization = **73.79%**

(c) Interpretation

- n = 6 provides better CPU utilization.
- More processes keep the CPU busy while others wait for I/O.

Question 6: CPU Scheduling

Given

Process Arrival Burst

P1	0	8
P2	0.4	4
P3	1	1

(a) FCFS

Gantt Chart

0 8 12 13

| P1 | P2 | P3 |

Completion Time

$$P1 = 8$$

$$P2 = 12$$

$$P3 = 13$$

Turnaround

$$P1 = 8$$

$$P2 = 12 - 0.4 = 11.6$$

$$P3 = 13 - 1 = 12$$

Average

$$\frac{8+11.6+12}{3} = 10.53 \text{ ms}$$

(b) SJF (Non-preemptive)

At time 0 only P1 has arrived.

So scheduling remains

$$P1 \rightarrow P2 \rightarrow P3$$

Average Turnaround Time

10.53 ms

(c) If Scheduler Knows Future Arrivals

Order

$$P3 \rightarrow P2 \rightarrow P1$$

Completion

$$P3 = 2$$

$$P2 = 6$$

$$P1 = 14$$

Turnaround

$$P3 = 1$$

$$P2 = 5.6$$

$$P1 = 14$$

Average

$$\frac{14+5.6+1}{3} = 6.87 \text{ ms}$$

Comparison

Algorithm	Average TAT
FCFS	10.53 ms
SJF	10.53 ms
Ideal SJF (Future Known)	6.87 ms

Question 8: Airport Check-in Scheduling

Since arrival times, service times, and passenger priorities are **not provided**, exact waiting time and turnaround time **cannot be calculated**.

General Comparison

FCFS

- Passengers served in arrival order.
- Simple and fair.
- Special-assistance passengers may wait longer.

Priority Scheduling

- Special-assistance passengers served first.
- Reduced waiting time for high-priority passengers.
- Better service efficiency during peak hours.

Conclusion

Priority Scheduling provides better service efficiency because urgent passengers receive faster service, though low-priority passengers may experience longer waiting times.